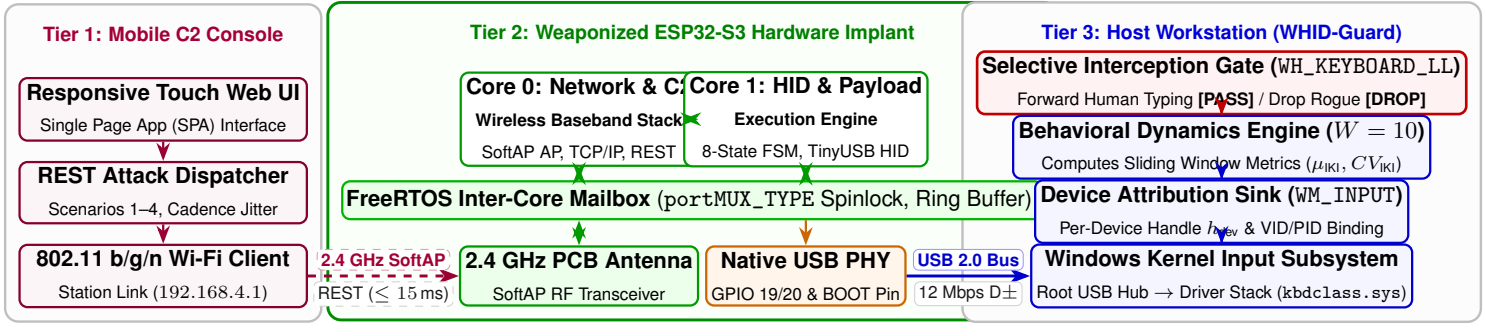

Tier 1: Mobile C2 Console
Responsive Touch Web UI
Single Page App (SPA) Interface
REST Attack Dispatcher
Scenarios 1–4, Cadence Jitter
802.11 b/g/n Wi-Fi Client
Station Link (192.168.4.1)
2.4 GHz SoftAP
REST (≤ 15 ms)
Tier 2: Weaponized ESP32-S3 Hardware Implant
Core 0: Network & C2
Wireless Baseband Stack
SoftAP AP, TCP/IP, REST
Core 1: HID & Payload
Execution Engine
8-State FSM, TinyUSB HID
FreeRTOS Inter-Core Mailbox (portMUX_TYPE Spinlock, Ring Buffer)
2.4 GHz PCB Antenna
SoftAP RF Transceiver
Native USB PHY
GPIO 19/20 & BOOT Pin
USB 2.0 Bus
12 Mbps D±
Tier 3: Host Workstation (WHID-Guard)
Selective Interception Gate (WH_KEYBOARD_LL)
Forward Human Typing [PASS] / Drop Rogue [DROP]
Behavioral Dynamics Engine (W = 10)
Computes Sliding Window Metrics (μIKI, CVIKI)
Device Attribution Sink (WM_INPUT)
Per-Device Handle hdev & VID/PID Binding
Windows Kernel Input Subsystem
Root USB Hub → Driver Stack (kbdclass.sys)